ModelOp

BMS CHIP Product Disposition Project Monitor Recommendations

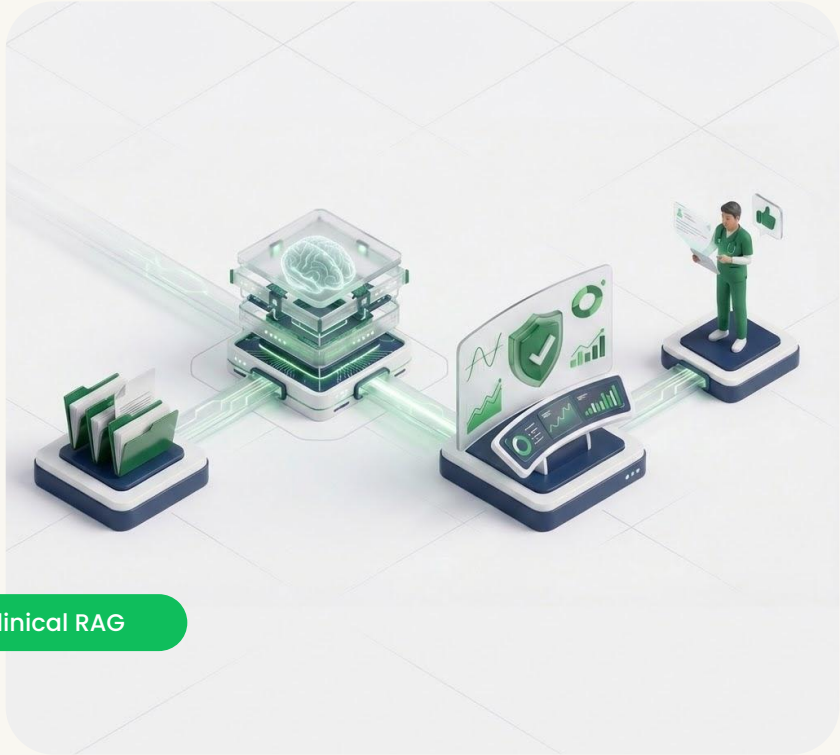
BMS CHIP Use Case: Product Disposition - Document (COA/Report) Validation
February 2026

Meeting Objective

ModelOp to provide the BMS CHIPs team monitoring recommendations for the “Documents (Reports) Validation” business use case.

Agenda

1. **Review use case:** ensure ModelOp understands business use case.
2. **Monitor & Report Recommendations:** based on use case success criteria.
3. **Discuss & Next Steps:** your success is ours as well



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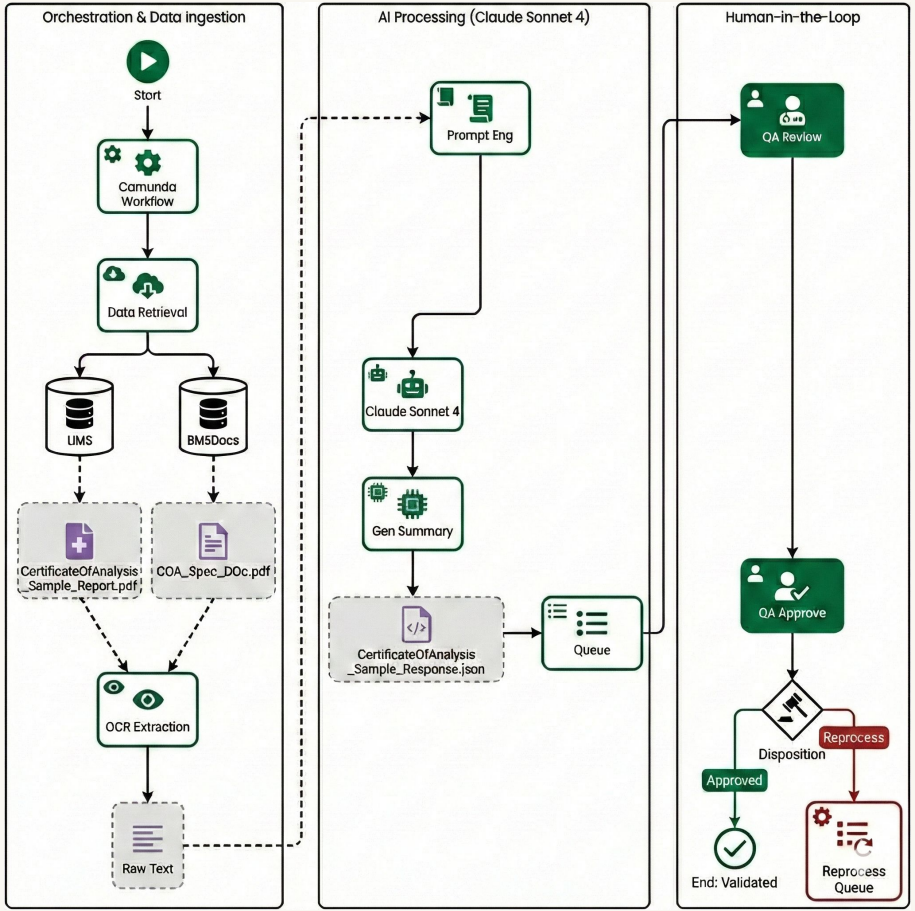
Process Flow

The Business Challenge:

- Manual review of *Certificates of Analysis (COA)* against *Specification Docs* is time-consuming and prone to human fatigue.
- **Goal:** Use AI to pre-validate "Conforms / Does Not Conform" statuses, allowing QA to focus only on exceptions.

The Solution's Implementation Architecture:

- **Input:** System automatically fetches PDFs (COA & Spec) from LIMS and BMSDocs.
- **Models:**
  a. **Preprocessing:** OCR/PDF Plumber extracts raw text/structured data.
  b. **Claude Sonnet 4** (AWS Bedrock) extracts data and applies logic.
- **Humans-in-the-Loop (HITL):**
  a. **QA Reviewer:** Reviews the document and LLM summary, checks against BMS policy.
  b. **QA Approver:** Makes final "validated" vs "reprocess" decision.



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Use Case Overview

- **Spec is the Goalpost:**
  The Spec document is the official definition of "perfection" *(conformance)* for cell therapy drug products

- **COA is the Scorecard:**
  The COA checks if a specific manufactured batch hits that perfection. Batches failing the Spec are blocked *("can't be released")*

- **Risk - Shifting Goalposts:**
  The Spec is dynamic *(7 versions so far, per Document History)*, meaning the "passing criteria" change. Automated systems (LLMs) must track the correct Spec version for each COA to avoid flagging a valid batch or, worse, passing a non-conformant one against old rules.

- **Monitoring Success Criteria:**
  For this to work, we need High Accuracy and High Agreement with human QA *(>95% concordance is the goal)*

  If the LLM misses identification of a deviation or incorrectly identifies a deviation *(false negative)*, a non-conformant product could slip through.

The core goal is monitoring that the LLM is accurately and consistently applying the dynamic Spec to every COA

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Recommended Monitors

Monitor 1 - Model Output Stability (Drift) Monitor:

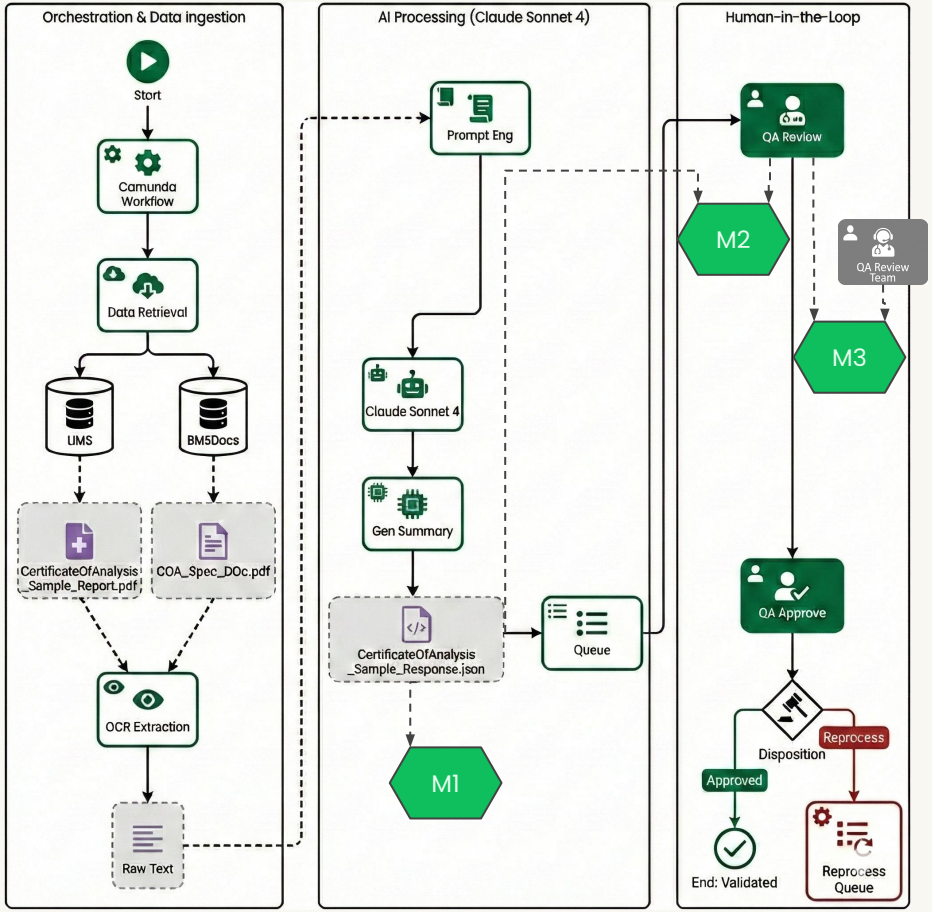
- Compares rejection rates this week vs. last. If it spikes without reason, alert.
- AI's historical baseline decisions vs. its current production decisions
- **Measures:**
  - Model Stability (PSI)
  - Reliability

Monitor 2 - Approval Concordance Monitor:

- Compare AI "Conform" vs. Human "Approve". If they disagree >5% of the time, flag it.
- LLM's Validation Recommendation vs. the Human QA's Final Vote
- **Measures:**
  - Concordance (Model-to-Human Agreement)
  - Accuracy

Monitor 3 - QA Calibration Monitor:

- Track Human QA rejection rates to ensure consistent policy application.
- QA Team's Previous Month's Rejection Rate vs. their Current Month's Rejection Rate
- **Measures:**
  - Human Stability (PSI)
  - Process Control



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Mapping BMS Goals to ModelOp Monitors

Monitor	BMS Goal	Description	Measures
1. Model Output Stability (Drift)	**Reliability** *(Model Stability)*	**Consistent Behavior:** The AI's rejection rate remains stable over time (PSI < 0.1) unless specific intervention occurs.	**"Is the AI getting moody?"** • We compare the AI's rejection rate this week vs. last month. • If it suddenly starts rejecting 50% more documents without a reason, we raise an alert.
2. Approval Concordance Monitor	**Accuracy & Agreement** *(Model-to-Human)*	**>95% Agreement:** High concordance between the LLM's recommendation and the Human QA's final decision.	**"Does the AI agree with the Human?"** • We compare the AI's 'Conform' vote against the QA's 'Approve' vote. • If the AI says Pass but the Human says Fail, we flag it.
3. QA Calibration Monitor	**Process Control** *(Human Stability)*	**Stable Review Standards:** Human QA rejection rates remain consistent over time, ensuring no 'policy drift' or fatigue.	**"Are the humans aligned with rules and each other?"** • We track the Human QA team's rejection rate over time. • If the humans suddenly get much stricter or more lenient than usual, we flag it for process review."

Reporting

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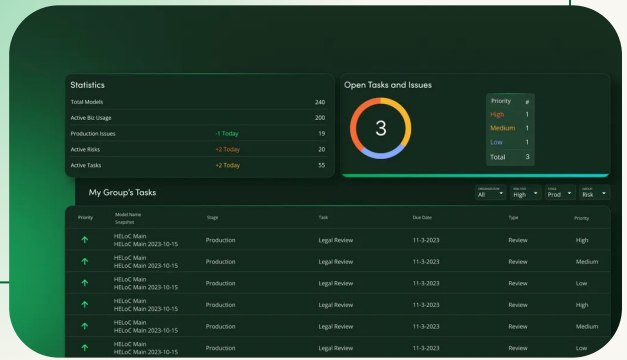

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Next Steps

Summary of Asks for BMS CHIPs Team

1. **Data:** We need historical logs or a "Reference Set" of available processed JSONs that include the Human Final Disposition. We will need to confirm schema of monitored data source(s)

2. **Access:** Confirmation of stakeholder MOC access to monitored endpoints within the full solution's context (i.e. Prompt and Claude Response).

3. **Config:** Confirmation of monitor granularity.
   Which LOD is more valuable: tracking the "Overall Status" = PASS or FAIL or Total Tests (aka Conform Votes).

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Thank you

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